

SRP3 — Sport Rehabilitation Practice 3

Evidence-Based End Stage Rehabilitation & Return to Play

Comprehensive Revision Notes | UZYKGA-30-3 | UWE Bristol



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SECTION 0 — Assessment Brief & Writing Framework

What the assignment requires

Core Task

- 2,000 words (exclusive of references) — UWE Harvard format
- You will be given ONE pathology (case study) — focus on END STAGE rehabilitation
- Choose and justify 4 exercises / drills appropriate for that pathology
- Identify and justify outcome measures used to evaluate readiness to RTP
- Discuss other factors affecting RTP (psychology, re-injury risk, MDT)

Recommended essay structure

Section	Content & Word Count
Introduction (~150w)	Brief background on pathology; define end-stage rehab; state your aim
Drill 1 (~250w)	Name, description, rationale, link to pathology stage, evidence citation
Drill 2 (~250w)	Same structure
Drill 3 (~250w)	Same structure
Drill 4 (~250w)	Same structure
Outcome Measures (~400w)	Choose 2–3; explain what each measures; justify selection for this pathology
Other RTP Factors (~350w)	Psychological readiness, re-injury risk, MDT decision-making
Conclusion (~100w)	Summary of programme rationale; criteria-based vs time-based approach

★ How to achieve high marks

- Critical evaluation: don't just describe drills — justify WHY with evidence
- Use peer-reviewed primary sources (journals > textbooks > websites)
- Show awareness of uncertainty: acknowledge limitations of evidence
- Integrate criteria-based RTP philosophy throughout (not just at the end)
- Demonstrate clinical reasoning — link each choice back to the stage of recovery

SECTION 1 — Mid-Portion Achilles Tendinopathy

Definition & Pathophysiology

From your notes

- A clinical syndrome characterised by localised tendon pain and swelling
- Involves failed healthy healing response and collagen disorganisation
- Rather than classic inflammation — this is key: it is a **DEGENERATIVE**, not inflammatory, process
- Located 2–6 cm proximal to the calcaneal insertion (mid-portion)

Additional pathophysiology to know:

- Neovascularisation (ingrowth of new blood vessels) accompanies the degenerative process
- Tendon loses its normal parallel collagen fibre arrangement → reduced tensile strength
- The tendon becomes thickened but mechanically inferior

Populations & Risk Factors

From your notes

- Basketball players / volleyball players / high jumpers (repetitive SSC loading)
- Sudden spike in plyometric volume or training load
- Limited ankle dorsiflexion
- ↑ Calf muscle weakness

- **ADDITIONAL:** Running on hard surfaces, hills, or sudden change in footwear
- **ADDITIONAL:** Higher BMI increases compressive load on tendon
- **ADDITIONAL:** Previous tendon injury is a risk factor for recurrence

Symptoms

- 2–6 cm proximal to calcaneal insertion — localised pain on palpation
- "Morning stiffness" — eases with movement
- "Warm-up effect" — pain **DECREASES** during exercise but **WORSENS** significantly the **NEXT MORNING**
- Tendon thickening visible/palpable

Clinical Signs & Tests

Test	Positive Finding & Significance
Palpation	Localised tenderness + tendon thickening at 2–6 cm above insertion
Single-Leg Heel Raise	Pain or reduced endurance during repetition — key functional test
Royal London Hospital Test ★	Compression in plantarflexion → pain; dorsiflexion → pain REDUCES = mid-portion (NOT insertional). Important to differentiate!

VISA-A questionnaire	Patient-reported outcome; score <80 indicates symptomatic; target >80 for RTP
Countermovement Jump / RSI	Assesses reactive strength & tendon energy storage-release (plyometric readiness)

Investigations

- Primarily a CLINICAL diagnosis
- Ultrasound (US) — shows tendon thickening, hypoechoic areas, neovascularisation
- MRI — more detailed if surgical planning or to exclude rupture

Atypical Presentations (Differentials)

- Achilles tendon RUPTURE — sudden onset, Thompson test positive, unable to weight bear
- Haglund's deformity — bony prominence at posterior calcaneus
- Retrocalcaneal bursitis — pain at insertion, not mid-portion
- Systemic inflammatory conditions (e.g. rheumatoid arthritis, spondyloarthropathy)

Prognosis

- Generally favourable with 12+ weeks of structured loading
- Jumping athletes have HIGHER RISK of recurrence if RTP is rushed before restoring tendon stiffness
- Criteria-based clearance is safer than time-based alone

★ COOK & PURDAM TENDON LOAD MODEL (Key Exam Content)

- Stage 1 — Reactive Tendinopathy: acute overload, tendon swells. Rx: REDUCE load, isometric holds for pain relief
- Stage 2 — Tendon Dysrepair: failed healing, early degenerative changes. Rx: controlled heavy slow resistance
- Stage 3 — Degenerative Tendinopathy: established degeneration, matrix disorganised. Rx: progressive loading, accept some irreversible change
- CRITICAL CONCEPT: Tendons are sensitive to COMPRESSIVE LOAD — avoid positions of ankle dorsiflexion under load, downhill running, and aggressive stretching in early stages
- Loading progression: Isometric → Heavy Slow Resistance (HSR) → Slow SSC → Fast SSC → Sport-Specific

End Stage Rehab — 4 Recommended Drills

These drills are appropriate for the end stage (return to sport phase):

Drill	Rationale & Evidence
1. Heavy Slow Resistance (HSR) Calf Raises (e.g. seated + standing heel raise with load)	Alfredson protocol adapted; HSR shown to remodel tendon collagen & improve mechanical properties. Eccentric + concentric loading. Targets both gastrocnemius and soleus.

2. Single-Leg Plyometric Hops / Pogo Hops	Progresses SSC loading; develops tendon stiffness and energy storage-release; assess with RSI (Countermovement Jump Test). Introduce when pain VAS <2/10
3. Running Progression (e.g. interval / tempo runs)	Sport-specific load; graduated increase in speed and volume. Monitor 24h pain response. Target: no increase >2/10 VAS post-session
4. Sport-Specific Jumping & Landing (e.g. box jumps, volleyball spike approach)	Replicates sport demands; fast SSC loading; final gate before RTP. Requires RSI symmetry >90%

RTP Criteria — Achilles Tendinopathy ★

4 Domains (from your notes — KEY EXAM CONTENT)

- PAIN: VAS < 2/10 during and after activity; no excessive next-day soreness
- STRENGTH: Single-leg weight-bearing capacity / calf endurance > 90% of contralateral side
- PLYOMETRICS: RSI symmetry > 90% on Countermovement Jump Test; able to perform sport-specific jumping tasks
- SPECIALIST SIGN-OFF: No more symptoms; cleared by physiotherapist ± sports medicine physician

Outcome Measures

Measure	What it assesses & RTP threshold
VISA-A (Victorian Institute of Sport Assessment - Achilles)	Patient-reported; 10 items; max 100. Score <80 = symptomatic. Target ≥80 for RTP. Most widely used PRO for Achilles tendinopathy
Single-Leg Heel Raise Endurance Test	Counts repetitions until fatigue/pain. Compares limb symmetry index (LSI). Target: ≥90% of contralateral side. Simple, equipment-free, direct functional relevance
Countermovement Jump Test (RSI)	Force plate or jump mat; measures reactive strength index = jump height ÷ contact time. Assesses plyometric readiness & tendon energy storage. Target LSI ≥90%

SECTION 2 — ACL Injury

Definition & Function

From your notes

- Partial or complete tear of the ACL — one of the primary stabilising ligaments of the knee
- Main function: prevent anterior translation of tibia relative to femur
- Also provides rotational stability

Populations & Risk Factors

- Demographics: Athletes — soccer, basketball, skiing, football
- Mechanisms: Non-contact pivoting; sudden deceleration; awkward landing
- Risk Factors: High Q-angle (biomechanical); weak core/gluteal
- ADDITIONAL: Female athletes have 2–8× higher ACL injury rate (valgus collapse, hormonal factors)
- ADDITIONAL: Previous ACL injury — 15× higher risk of re-rupture

Symptoms

- Acute: "Pop" sound/sensation at time of injury
- Immediate haemarthrosis (blood in joint) → significant swelling within hours
- Inability to continue playing
- Chronic: "Giving way" / rotational instability during pivoting

Clinical Signs & Tests

Test	Description & Significance
Lachman Test ★	Most sensitive & specific for ACL; anterior tibial translation at 30° flexion. Soft end-feel = positive
Anterior Drawer Test	Anterior translation at 90° flexion; less sensitive than Lachman but widely known
Pivot Shift Test	Tests rotational instability; most specific for functional deficit; difficult to perform acutely

Investigations

- X-Ray: exclude fractures (Second fracture associated with ACL injury)
- MRI: Gold standard — confirms diagnosis, identifies associated meniscal/cartilage damage
- CT Scan: bone loss assessment pre-surgical planning

Atypical Presentations

- "Dead Arm" equivalent: sudden paralysing pain during a throw — note: this is more shoulder-specific from your notes; for ACL, atypical = minimal pain but significant instability

- Multi-ligament injury: ACL + PCL + collaterals — complex instability pattern
- Combined meniscal tears — may mask or worsen symptoms

Prognosis

- Age factor: recurrence rates are highly age-dependent (younger = higher re-injury risk)
- Surgical: High success rates 80–95% for traumatic cases (Bankart — note: from shoulder notes; ACL surgery = ACLR with graft)
- ACLR Graft options: Bone-patellar tendon-bone (gold standard), hamstring autograft, quadriceps tendon
- Conservative management possible for lower-demand individuals
- Return to cutting/pivoting sport typically 9–12 months post-ACLR

End Stage Rehab — 4 Drills (from your notes) ★

Drill	Rationale
1. Spanish Squat	Isometric/isotonic quad strengthening with reduced tibial shear force; targets VMO; good for reloading post-ACLR without anterior tibial stress. Knees track over toes with band resistance.
2. Nordic Hamstring Curl	Gold standard eccentric hamstring exercise (Level 1 evidence for hamstring injury prevention); improves H:Q ratio; targets semitendinosus which is key for rotational stability post-ACLR
3. Single-Leg Landing with Hold	Neuromuscular control & landing mechanics training; addresses valgus collapse; requires knee stability and proprioception; assesses movement quality before progressing to dynamic tasks
4. Figure-8 Run	Functional change-of-direction task; replicates sport-specific demands; tests confidence and neuromuscular control in multi-directional cutting; used as a performance/readiness test

RTP Criteria — ACL

Criteria-Based RTP (not just time)
• Quadriceps LSI $\geq 90\%$ (isokinetic or single-leg hop battery) • H:Q ratio $\geq 66\%$ at high speed (isokinetic testing) • Single-leg hop test battery LSI $\geq 90\%$: single hop, triple hop, crossover hop, 6m timed hop • ACL-RSI (psychological readiness scale) — score should be adequate (>56 considered threshold) • Minimum 9 months post-ACLR (time + criteria combined) • Clearance from surgeon + physiotherapist (MDT)

Outcome Measures — ACL

Measure	What it assesses & RTP threshold
Limb Symmetry Index (LSI) via Single-Leg Hop Battery	Compares injured vs uninjured limb; $\geq 90\%$ across all 4 hop tests = functional RTP criterion. Most widely used functional battery
IKDC (International Knee Documentation Committee)	Patient-reported; subjective knee function; score out of 100; higher = better function

ACL-RSI (Return to Sport after Injury Scale) ★	Psychological readiness; 12 items; addresses emotions, confidence, risk appraisal. Low scores predict re-injury. Target >56
Isokinetic Strength Testing	Quad & hamstring peak torque; H:Q ratio; LSI thresholds. Requires Biodex/isokinetic dynamometer

SECTION 3 — Shoulder Instability

Definition

From your notes

- Clinical condition where the humeral head does not remain centred in the glenoid fossa during shoulder movement
- This leads to excessive translation of the humeral head, resulting in pain, partial sliding out, or dislocation

Populations & Risk Factors

- Young males around 15–30 years old
- Overhead athletes & contact sports
- PREVIOUS DISLOCATION is the strongest predictor of future instability ★ (from your notes — key exam point)
- Anatomy: Glenoid dysplasia (shallow glenoid = less bony stability)
- ADDITIONAL: Hyperlaxity / generalised joint hypermobility
- ADDITIONAL: Bankart lesion (labral avulsion from anterior glenoid rim)
- ADDITIONAL: Hill-Sachs lesion (posterior humeral head compression fracture)

Symptoms

- Deep shoulder pain of posterior aching character
- Especially in the "apprehension position" (90° abduction, ER)
- Psychological fear of feeling the arm will "come out the socket"
- Clicking, clunking, or giving way sensation

Clinical Signs & Tests

Test	Description & Positive Finding
Apprehension Test	Arm in 90° abduction + external rotation → patient reports fear/apprehension of dislocation
Relocation Test ★ (from your notes)	Posterior force applied to humeral head RELIEVES apprehension = positive. Confirms anterior instability
Anterior Drawer Test	Anterior translation of humeral head in glenoid; compare bilaterally
Sulcus Sign	Inferior distraction produces sulcus below acromion; indicates multidirectional instability (MDI)
Load and Shift Test	Manually loads humeral head then shifts anterior/posterior; grades translation

Investigations

- X-Ray: Hill-Sachs lesion, glenoid bone loss, fractures
- MRI / MR Arthrogram: Bankart lesion, SLAP tear, labral pathology

- CT Scan: Quantifies glenoid bone loss (critical for surgical planning — Latarjet threshold ~20–25% loss)
- Useful resources noted: ATOS → likely ASES (American Shoulder & Elbow Surgeons) score; Orthopaedic surgeons

Atypical Presentations

"Dead Arm" Syndrome ★ (from your notes)

- Sudden paralysing pain and weakness in athlete's arm during a throw
- Often WITHOUT a clear dislocation event
- Likely transient neurological compromise (brachial plexus traction) associated with subluxation
- Important to distinguish from true cervical/neural pathology

Prognosis

- Age factor: Recurrence rates are HIGHLY age-dependent (under 20 = up to 90% recurrence conservatively)
- Surgical: 80–95% success for traumatic cases with localised labral tear requiring repair
- Athletes in contact sports: higher recurrence risk; earlier surgical consideration

Management

- Conservative (physiotherapy): Strengthening of the ROTATOR CUFF (subscapularis, infraspinatus, teres minor, supraspinatus) + scapular stabilisers
- Surgical — Arthroscopic Bankart Repair: for labral tears (anterior instability); good outcomes for first-time traumatic dislocators
- Surgical — Open Latarjet Procedure: for significant bone loss (>20–25% glenoid); transfers coracoid process to anterior glenoid to restore bony stability

End Stage Rehab — Example Drills for Shoulder Instability

Drill	Rationale
1. Rotator Cuff PRE (e.g. ER/IR cable exercises)	Targets infraspinatus/teres minor (ER) and subscapularis (IR); improves dynamic stabilisation of humeral head; evidence supports progression from isometric to isotonic to functional
2. Perturbation Training (e.g. push-up on unstable surface)	Challenges proprioception and neuromuscular control of the shoulder complex; improves reflex stabilisation; appropriate for late-stage rehab
3. Plyometric Throwing Progression (e.g. wall throws, partner drills)	Sport-specific; gradually reloads the shoulder through throwing mechanics; progresses velocity and load; functional pre-RTP task
4. Sport-Specific Overhead Task (e.g. overhead press, spike, serve)	Final functional gate; replicates sport demands in apprehension position; should be pain-free with no apprehension

Outcome Measures — Shoulder Instability

Measure	Description & use
WOSI (Western Ontario Shoulder Instability Index) ★	Most specific PRO for shoulder instability; 21 items across 4 domains; score /2100 (lower = better, sometimes converted to %)
ASES (American Shoulder & Elbow Surgeons Score)	50% pain VAS + 50% function questions; score /100; widely used in research and clinical practice
Isokinetic ER/IR Strength Ratio	Target ER:IR ratio ~66%; LSI >90%; demonstrates rotator cuff symmetry for clearance
Apprehension-free functional test	Clinical sign-off criterion — patient must perform sport-specific movement without apprehension

SECTION 4 — Hamstring Injury

Definition & Anatomy

- Strain or tear of the hamstring muscle group (biceps femoris, semitendinosus, semimembranosus)
- Graded I–III (mild strain → complete rupture)
- Most common in sprinting athletes — occurs during the late swing phase (maximum eccentric load)
- Proximal hamstring avulsions are a specific subtype

Populations & Risk Factors

- Sprinting sports: soccer, rugby, athletics, AFL
- Previous hamstring injury is the SINGLE STRONGEST predictor of recurrence (20–30% re-injury rate)
- Fatigue, poor warm-up, rapid acceleration
- Imbalanced H:Q ratio
- Age — older athletes at higher risk

Clinical Tests

- Asking H-test: differentiates Type I (sprinting) vs Type II (stretching) mechanism — affects prognosis
- Straight Leg Raise (SLR) with pain: neurodynamic vs hamstring
- Active Knee Extension: quantifies range

End Stage Rehab — Key Drills ★

From your notes — the 2 key exercises

- A-SKIP: High knee + rhythmic coordination; trains the forward swing phase; improves hip flexion mechanics and running gait. Common in end-stage running rehab.
- B-SKIP: Adds a leg extension/pawback to A-skip; mimics the late swing → foot strike phase; trains the exact movement where hamstrings are under peak eccentric load.
- WHY these matter: They systematically rebuild running mechanics before full sprint exposure — a key principle in hamstring rehab progression.

Drill	Rationale
1. Nordic Hamstring Curl	Gold standard eccentric exercise; most evidence-based hamstring injury prevention exercise (Level 1a); reduces re-injury risk by ~50% when used in prevention programmes. Essential in end-stage.
2. Spanish Squat (if needed)	Can supplement — but primarily for quads. For hamstrings, Nordic + RDL are preferred.
3. A-Skip & B-Skip (running mechanics) ★	Systematically rebuilds sprint mechanics; low risk; high specificity for the injury mechanism (late swing eccentric load). Progress from A-skip → B-skip → build-up runs → max sprint.

4. Progressive Sprint Runs (e.g. 50% → 70% → 90% → 100% max velocity)	Final functional gate; must reach ≥95% of pre-injury sprint speed; GPS/timing gates used to measure. Cannot safely skip this step before RTP.
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RTP Criteria — Hamstring

Key RTP gates

- Pain-free during ALL drills and sprint efforts
- Isokinetic H:Q ratio ≥ 66% (or match contralateral side)
- Sprint speed ≥ 95% of pre-injury or contralateral benchmark (via GPS / timing gates)
- Asking tests and SLR: symptom-free
- Psychological readiness: no fear of sprinting
- SCAT-6 note: primarily a concussion assessment tool — may have been mentioned in context of contact mechanisms (e.g. tackle causing both head and hamstring injury simultaneously). Not a primary hamstring-specific tool.

Outcome Measures — Hamstring

Measure	Description
Isokinetic Testing (H:Q ratio)	Target H:Q ≥ 66%; LSI ≥ 90%. Biodex required. Most objective measure of muscle imbalance.
Sprint Speed Testing (GPS / timing gates)	≥95% of max velocity reached. Most sport-specific and directly relevant to injury mechanism.
HAGOS (Hip and Groin Outcome Score)	If proximal involvement; 6 subscales. Not always used but worth knowing.
Asking H-test	Clinical differentiation of mechanism type; guides prognosis and rehab design

SECTION 5 — Universal RTP Framework & Other Factors

Core Philosophy: Criteria-Based > Time-Based RTP ★

- Modern RTP decision-making is based on meeting functional, psychological, and clinical CRITERIA — not simply waiting for a set number of weeks
- This should be stated in your assignment introduction and conclusion

Other Factors Affecting RTP (Essay Section ~350w)

1. Psychological Readiness

- Kinesiophobia (fear of movement/re-injury): measured by Tampa Scale for Kinesiophobia (TSK-11/17)
- ACL-RSI for knee injuries (12 items; addresses emotions, confidence, risk appraisal)
- Athletes with low psychological readiness have SIGNIFICANTLY higher re-injury rates
- Psychological readiness is often the last criterion to be met, even when physical criteria are satisfied
- Strategies: graduated exposure, mental imagery, CBT-informed approaches

2. Re-injury Risk

- Previous injury = strongest predictor for most pathologies (ACL, hamstring, shoulder)
- Sport type: contact / high-impact sport carries higher re-injury risk
- Limb symmetry not fully restored = biomechanical compensation patterns
- Age: younger athletes (ACL) have higher re-rupture rates
- Training load management post-RTP: gradual reintegration to full training

3. Multidisciplinary Team (MDT) Decision

- RTP is NOT a unilateral physiotherapist decision
- MDT: Sports medicine physician, physiotherapist, strength & conditioning coach, coach, athlete
- All criteria must be met AND team agreement reached
- Athlete's own readiness and consent is an ethical requirement

4. Sport & Competition Demands

- Contact vs non-contact: higher risk tolerance may be needed for contact sports
- Position-specific demands: goalkeeper vs forward have different loading profiles
- Competition calendar: external pressure to rush RTP must be resisted
- Training load monitoring: GPS, session-RPE, ACWR (acute:chronic workload ratio) post-RTP

5. Psychological / Social Factors

- Fear of failure, team pressure, financial pressures (professional athletes)
- Family/coach expectations — may conflict with medical advice
- Self-efficacy: athlete's belief in their ability to perform at pre-injury level

SECTION 6 — Outcome Measures Quick Reference

Outcome Measure	Pathology & Key Details
VISA-A	Achilles Tendinopathy Patient-reported 0–100 Target ≥80 for RTP
Single-Leg Heel Raise Endurance	Achilles Count reps to fatigue Target LSI ≥90%
Countermovement Jump / RSI	Achilles / ACL / Hamstring Force plate or jump mat RSI = height ÷ contact time LSI ≥90%
IKDC	ACL Patient-reported 0–100 Higher = better knee function
ACL-RSI	ACL Psychological readiness 0–100 Target >56 12 items
Single-Leg Hop Battery (LSI)	ACL / Hamstring 4 tests ≥90% LSI for RTP Most validated functional battery
Isokinetic Strength Testing	ACL / Hamstring / Shoulder H:Q ratio ≥66%; LSI ≥90%; Biodex required
WOSI	Shoulder Instability 21 items /2100 lower = better Most specific for instability
ASES	Shoulder 0–100 Pain + function subscales Widely used in research
Tampa Scale (TSK)	ALL pathologies Kinesiophobia 11–17 items High scores = poor psychological readiness
HAGOS	Hamstring (proximal) / Hip 6 subscales Hip and groin function
Sprint Speed (GPS / Timing Gates)	Hamstring / ACL ≥95% max velocity Most sport-specific hamstring RTP gate

Final Tips for the Assignment

- State your pathology clearly in the introduction and define "end stage rehabilitation"
- For each drill: name it → describe it → explain WHY (link to pathology physiology + stage) → cite evidence
- For outcome measures: name it → what does it measure → WHY is it appropriate for THIS pathology → what threshold indicates RTP readiness
- In the "other factors" section: bring in psychological readiness (specific scale), re-injury risk factors relevant to the pathology, and MDT
- Use the Cook & Purdam model (Achilles) or criteria-based RTP philosophy to show critical appreciation
- Avoid time-based language ("after 12 weeks") — use criteria-based language ("once LSI ≥90% is achieved")
- Reference peer-reviewed journals; avoid relying on textbooks or websites

Good luck — you've got this! 加油💪